

Nathaniel Christen
8331 Meadowsweet Road
Pikesville, MD 21208
(443) 547-2018
Case # 328806759

Personal Statement

During a phone interview on December 16, 2025, I was asked to compose a Personal Statement outlining my source of income and other relevant information to upload via the Maryland Benefits document portal.

I did not receive any specific instructions about the desired format for this document, so I have constructed it in paragraph form. I will briefly mention de facto “part time” work and then discuss longer-term projects relevant to my current housing and employment situations.

With respect to part-time “gigs”, shortly after moving to Maryland in May of 2025, I met a few neighbors who needed help with dog-walking (I myself had a pitbull mix in Brooklyn that had to be rehomed with a friend of the family before I left, and I’ve had experience in the past with dog walking/sitting). Currently I walk dogs for roughly 3–4 hours per day (every day) mostly from approximately 10:00 – 11:30, 4:15 – 5:15, and 6:00 – 7:00, occasionally with a couple other owners as well when they are out of town or exceptionally busy (for one owner I sometimes feed her birds and clean a bird cage). I haven’t established a fixed fee schedule, but in general I’ve received \$40 per week from one owner, and \$80 per month from another (the latter was affected by the government shutdown, such that I did not receive payment for two months around September and October), so in general I receive about \$240 each month. For the aforementioned owner with birds (and two small dogs I’ve walked on occasion) I might get an

additional \$20/month on average.¹ I sometimes also walk for a neighbor with a very charming chocolate lab, but that's not for monetary compensation.²

Several times I have heard from other neighbors that they too need dog walking services (or cat-sitting on one occasion), but never got contacted back. It seems difficult to make informal pet care one's primary source of income.

In sum, when the dust settles I can figure roughly \$200-\$250 per month working 3.5 – 4 hours a day. This helps with incidental expenses (food, mostly) but I need to provide a little backstory to address the situation with where I am living and related details.

Since roughly 2015 I have been employed³ by a small start-up based in New Jersey which is currently focused primary on publishing technology. Before the Covid pandemic we had also devoted attention to biomedical software.

I have a BA-level computer science degree, but I ended up pursuing a PhD in philosophy, specifically philosophy of science. Academically, I was especially interested in philosophy of science in conjunction with Cognitive Linguistics (both linguistics as a science and, conversely, applying cognitive-linguistic models to analyze the cognitive foundations of scientific reasoning).

Cotemporous with my doctoral studies (which were in Canada), I was helping with my parents' business, implementing an e-commerce platform.⁴ While rewarding — and profitable online — the “brick-and-mortar” business did not really pan out and my family ended up struggling (my father committed suicide; my mother had a stroke and needed live-in help). In the end, splitting time between the US and Canada, I had trouble renewing my Canadian visa and did not defend my dissertation, although I did complete a second version after getting

¹ Subtracting the government shutdown-related gap essentially cancels that out

² Rather, I join them afterward for dinner.

³ informally, at least

⁴ There was actually a science connection, because my father was trying to launch a store as an antiques collector, and much of his inventory was scientific instruments or other devices with patents, or other information we could investigate about how the tools or machines were constructed and used. I built a software tool that connected to the US Patent Office and helped us match patent images with our inventory items.

feedback for the initial draft from my committee. After my father died my mother and I continued the business for several years in solely-digital fashion, during which time we moved (back) to New York City. There I attended small-business seminars, tech meetup groups, and other programming-related activities, with the goal of potentially migrating the family business from our own e-commerce shop toward providing website-management software for other businesses. This was how I eventually connected with the startup mentioned earlier.

As a doctoral student I had at one point served on a committee that was redesigning our departmental website, with the goal of providing a publishing platform and search/annotation system for our students and faculty. From a philosophy of science perspective, I was interested in how “executable research objects” and other Science Grid technologies could help document theoretical and methodological foundations for scientific research.

The startup founder had similar interests; she was initially focused on Natural Language Processing, but also engaged with health-care applications.⁵ She had contacts in the government and we put together numerous slide decks and presentations for US “scouts” as well as hospitals and medical practices. Her government contacts encouraged us to develop software for “under-resourced” communities, and we eventually began collaborating with a startup that was New York based (supported by an NYU incubator program) but doing field work in India, deploying informatics and diagnostic tools to rural clinics.⁶ This led us to build front-end components in collaboration with the Indian group.

Our general paradigm was that open-source EMR components should be designed as embedded plugins, or via similar coding conventions and interoperability, to existing software focused on research, bioimaging, immuno-oncology, and other bioinformatics areas. This is in contrast to the way EMR systems are usually conceived, which are more isolated to the clinical setting. Our theory is that grid-computing techniques are conducive to EMR systems that are more

⁵ She had done volunteer work at an Israeli hospital, and several of her advisers were doctors. As a student I had also briefly worked at a refugee camp in Thailand.

⁶ Open-source Hospital Information Systems, Electronic Medical Records, and Operating Systems (such as BioLinux) can offer an important savings for hospitals and point-of-care providers, since there are no licensing fees.

adaptable (e.g., for hospitals that develop community-oriented initiatives, or require special forms for clinical trial management, etc.).

Unfortunately, the Covid pandemic made it difficult to continue unfunded projects in perpetuity, and the startup's founder decided to shift attention to technology for academic publishing. During this time I wrote an Elsevier book on biomedical software, plus eight chapters or articles on various computer-science topics (e.g., computational linguistics and compiler theory); she (the founder) was working on her own book contract and many op-eds and popular-media articles. In short, we both had multiple publishing projects and I found it worthwhile to implement new publishing software, even if (at that point) just for our own internal use.

In addition to science (as editor of a Springer Nature journal and a technology book series) the founder has been involved in political activism, particularly in the area of family-court reform. In this capacity she sometimes provides counseling to mothers engaged in custody battles, or restricted to visiting children as non-custodial parents. During the pandemic, we essentially became a small-scale publishing house, compiling legal documents, newsletters, graphics design (e.g., for real estate ads), translation, and other document-preparation tasks. In some cases we built custom publishing software to address requirements for individual projects, including producing two full-length books (eventually published by Elsevier and Oxford University Press respectively) and the equivalent of a third book as the dataset component of a Springer Nature article.

I received a book contract and some compensation for working on legal documents, graphics design, translation, etc., but on a day-to-day basis much of my work was centered on the founder's advocacy, maintaining a web site and preparing/preserving articles, op-eds, and other materials she published in venues including the New York Times and Washington Post, plus many smaller papers. In exchange, she arranged for me to stay with a colleague and paid rent on my behalf (probably the equivalent of about \$850/month on the open market). Recently my landlady sold her house, which led to my relocating to Baltimore under a similar arrangement with a close associate of the founder, who has coauthored with her and shares her interest in family-court reform.

Leveraging rent-free housing in exchange for “in kind” work isn’t a great long-term solution, so I hope SNAP benefits would allow me to reallocate funds I receive from more informal “gig” engagements toward a more secure rental situation. Meanwhile, instead of relying on one-off work on individual publishing projects we are building more substantial software for potential use either by publishing houses or curatorial organizations, such as PubMed Central and the National Institutes of Health. I have implemented numerous code prototypes and presentations about unique features of our software, which has been used to build full-scale books and research data sets.

Supplement

A few comments about science and publishing technology, for clarification.

Research networks or “grids” are an increasingly popular architecture for collaborative and decentralized science communities. Examples include the Cancer Bioinformatics Grid (caBIG), Center for the Development of a Virtual Tumor (CViT), National Cancer Informatics Program (NCIP), Data Infrastructure Building Blocks (from the National Science Foundation), Open Science Grid (OSG), Worldwide LHC Computing Grid (WLCG), EarthCube, Earth System Grid Federation (ESGF), and many others. Although originally designed to support decentralized parallel computing — that is, for original research — grids have evolved into pedagogical and reproduction tools as well as hosts for initial experiments.⁷

Grid specifications are often focused on metadata standards. There are not necessarily standard guidelines for actual *code*, as written in common science-computing languages, for components that will be hosted on a grid. I consider this a lacuna, because there are detailed features and annotations that may be applied to help grid components interoperate, and to increase overall usability, especially with an emphasis on FAIR (Findable, Accessible, Interoperable, Reusable) priorities.

⁷ “caGrid 1.0: An Enterprise Grid Infrastructure for Biomedical Research” (2008)
(<https://pmc.ncbi.nlm.nih.gov/articles/PMC2274794/>)

One rationale for grid computing is to provide remote services for investigations that require special-purpose equipment — capacities beyond that typically available to researchers. However, a different aspect of grid architecture is effectively the opposite: allowing scientists and students to replicate research (perhaps in a scaled-down version) on their own computers, or otherwise in quotidian computing environments. This serves both data-transparency and pedagogical goals (a concrete forum for fellow investigators to understand published results more thoroughly). Given these use-cases, it is important for executable research objects to be developed which do *not* need special-purpose hardware.

There are many advanced science applications that are largely self-contained: they can be compiled on ordinary desktop computers, with few external dependencies, and can form the basis of FAIR data sets. So, one may design executable research objects wherein data visualization/analysis tools are shared as source files internal to the data set, perhaps by including (in whole or in part) source-code archives for relevant discipline-specific applications, to be compiled on site (in lieu of downloading prebuilt binary executables). These data packages could likewise contain customized PDF viewers, and the relevant PDF and scientific software may be integrated together for indexing and cross-referencing.

The resulting Research Object model is based on code packages distributed as standalone source archives with minimal extra dependencies, rather than as binary components or as assets that require external software. This paradigm is different from how Research Objects are typically disseminated. Such differences inspire our startup to work on science-grid technology even though we have no direct affiliation with universities that participate in one or more grid networks, because we are adopting an approach not pursued elsewhere.

Not speaking for everyone involved with the startup, but I personally remain committed to the work we did earlier on clinical information systems and EMR software for under-resourced communities. There are technical similarities between source-focused grid packages and extensible EMR platforms, and I hope that concrete deployment of instances of the former kind can help us transition to implementation of components of the latter variety as well. Building EMR software via science-grid resources can help with migrating research results to on-site clinical practice, in accord with “Translational Medicine” initiatives.